



আন্তর্জাতিক ইসলামী বিশ্ববিদ্যালয় চট্টগ্রাম
الجامعة الإسلامية العالمية شيتاغونغ
International Islamic University Chittagong

PROJECT REPORT

HOSPITAL MANAGEMENT SYSTEM

Course Code: CSE-3532

Course Title: Tools and Technologies for Internet Programming

Submitted By:

Sharmin Akter Shorna

ID- C231443

Trishita Paul

ID- C231471

Submitted To:

Sara Karim

Adjunct Lecturer

Department of CSE, IIUC

Overview

SmartCare is an advanced hospital management system built using Django and Django REST Framework (DRF), designed to improve patient care and hospital efficiency. It offers seamless doctor-patient interaction, secure user authentication with email confirmation, appointment scheduling with automated email alerts, and real-time search and filtering for doctors based on name or specialization.

How it is useful

SmartCare is a useful hospital management system that simplifies hospital tasks like appointment booking, doctor schedules, and patient data handling through automation. It allows patients to easily search for doctors, book appointments, and receive real-time email notifications. Doctors can manage appointments through a dedicated panel, while the responsive design ensures smooth use on both desktop and mobile devices. With secure authentication, including email verification and token-based access, user data is kept safe.

Why This Project

SmartCare is important because it improves the overall patient experience by making healthcare more accessible and organized. It saves time for both hospital staff and doctors by reducing manual workload. All data is securely stored and managed centrally, ensuring easy access and privacy. By modernizing healthcare operations with scalable technology, SmartCare helps bring hospitals into the digital age efficiently and safely.

Technologies Used:

Backend

- **Django** (*Python Web Framework*) — powering the backend APIs, models, and building logic
- **Django REST Framework (DRF)** — crafting clean and scalable RESTful APIs for the frontend
- **SQLite** — lightweight and efficient data storage solution

Frontend

- **HTML, CSS, JavaScript** — building interactive, responsive, and user-friendly web pages

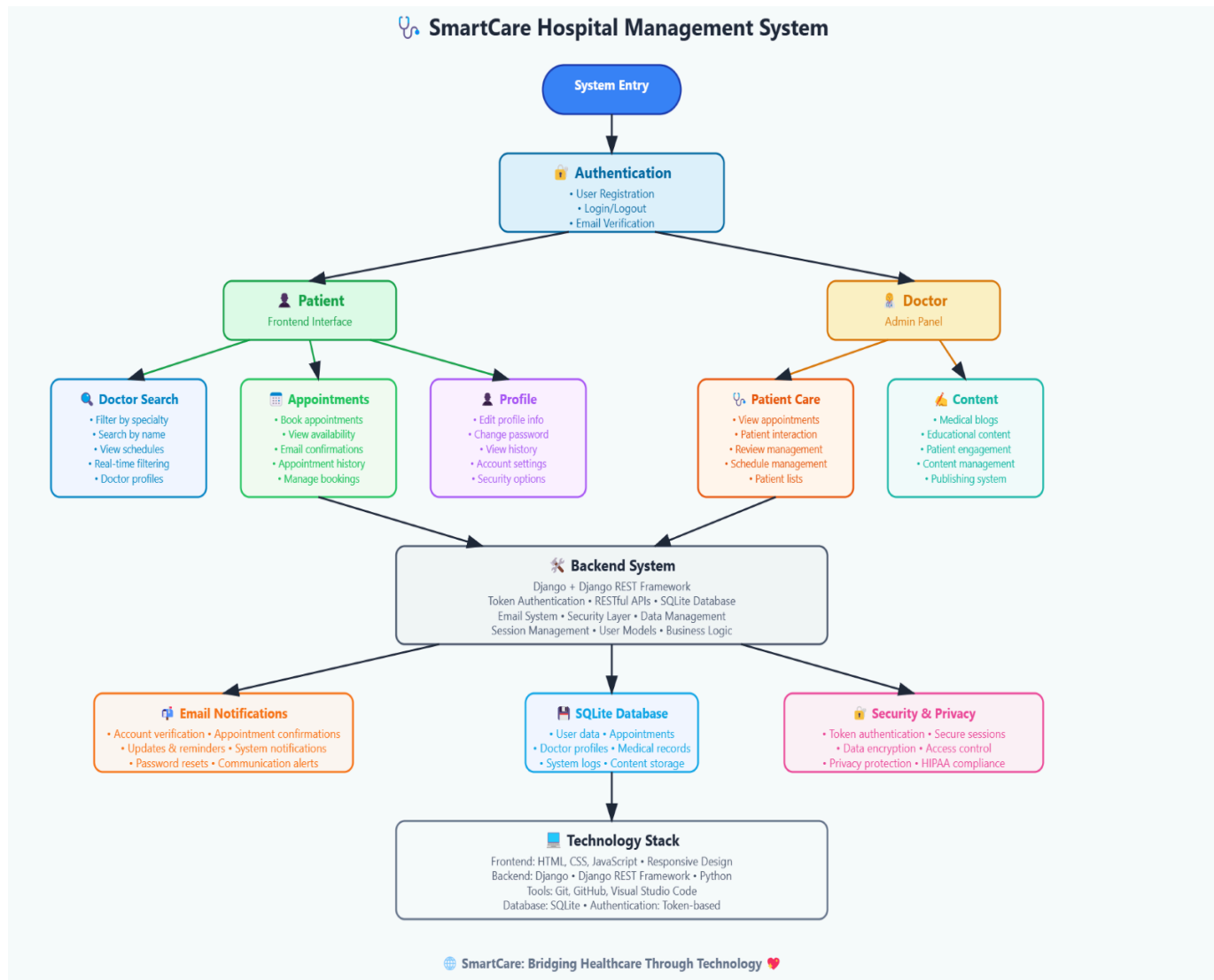
Authentication & Security

- **Django's built-in User model & authentication system** — robust and secure user management
- **Token Authentication (DRF)** — ensuring secure API access and user sessions
- **Email Verification System** — confirming user accounts through email to keep it safe and sound

Tools & Environment

- **Git & GitHub** — seamless version control and team collaboration
- **Visual Studio Code** — your cozy IDE for coding and debugging with love 

Flowchart



Design Process

Our project, **SmartCare - Hospital Management System**, was carefully designed and developed through the following steps to ensure a smooth, user-friendly, and efficient healthcare experience:

1. Requirement Gathering

We began by identifying the key needs of both patients and doctors — such as appointment booking, doctor search, profile management, and communication — to create a comprehensive feature list.

2. **System Design & Architecture**

We designed a modular system architecture separating the frontend and backend, using Django REST Framework for APIs and a responsive frontend built with HTML, CSS, and JavaScript for a clean UI.

3. **Frontend Development**

Created responsive web pages allowing users to search doctors, book appointments, manage profiles, and read blogs, with real-time API integration using JavaScript.

4. **Backend App & Model Setup**

The backend was initialized by creating Django apps for different modules (e.g., patients, doctors, appointments).

- Defined models for users, doctors, appointments, reviews, and blogs
- Created **serializers** to handle data conversion between models and APIs

5. **Backend Development**

Implemented RESTful APIs to handle authentication, appointment management, doctor profiles, and blog posting. Secure email verification and notification systems were integrated for smooth communication.

6. **Testing & Debugging**

Conducted thorough testing on both frontend and backend to ensure functionality, usability, and security. Bugs were fixed promptly to deliver a stable product.

7. **Deployment & Documentation**

Uploaded the complete project to GitHub with clear documentation, including README and project report, to assist users and future developers.

Future Plans & Next Phases

While **SmartCare** already offers essential features for hospital management, we have exciting ideas to expand and improve the system in future phases:

Phase 1: Enhanced User Experience

- **Live Chat Support** — Allow real-time chat between patients and doctors for faster consultations

- **Multi-language Support** — Make the platform accessible to a broader audience
- **Role-Based Dashboards** — Separate views for admin, patients, and doctors with personalized data
- **PDF Generation** — Allow users to download appointment confirmations and summaries in PDF format

Phase 2: Improved Security & Authentication

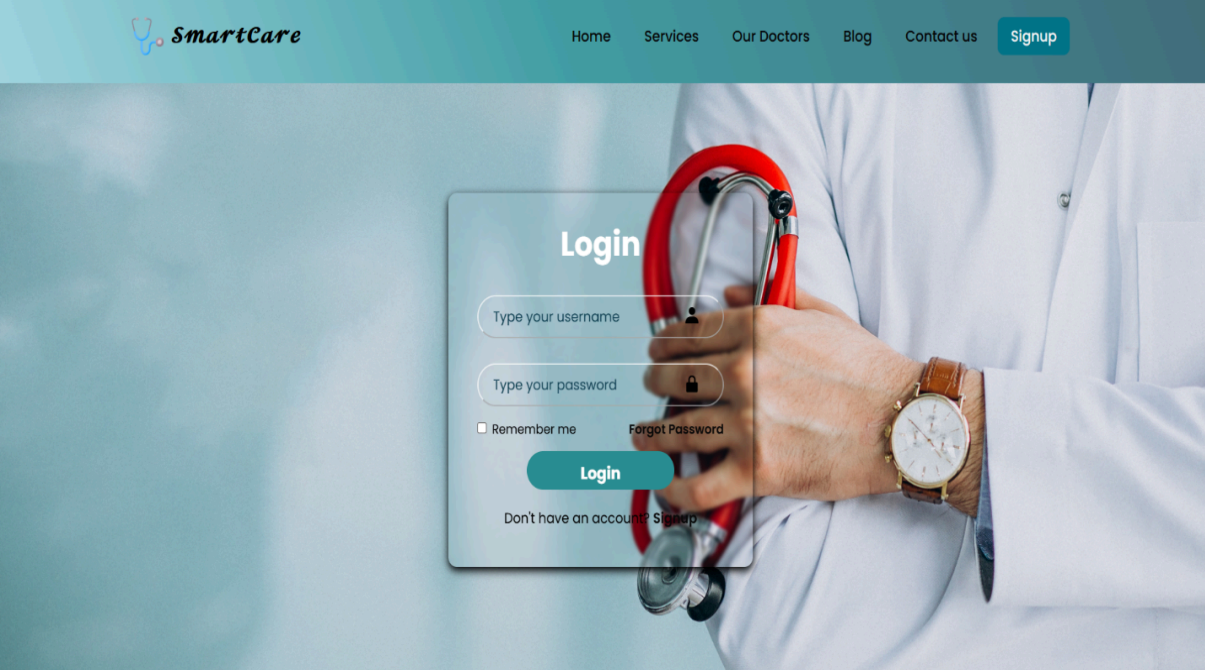
- **Two-Factor Authentication (2FA)** — For an added layer of account protection
- **Password Reset via Email/SMS** — To help users securely regain account access

Phase 3: Advanced Features & Automation

- **Analytics Dashboard** — Display patient visit stats, doctor performance, and service usage
- **AI-based Appointment Suggestions** — Recommend the best available doctors based on symptoms
- **E-Prescription Module** — Doctors can issue digital prescriptions viewable by patients

Phase 4: Mobile App Development

- **Android & iOS App** — A fully functional mobile version of SmartCare for on-the-go access
- **Push Notifications** — For reminders, confirmations, and real-time updates



Our Services

We provide top-notch healthcare services for your well-being.



Dental Surgery
Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the ...



X Ray
Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the ...



MRI
Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the ...



COVID 19 Test
Lorem Ipsum is simply d text of the printing a typesetting industry. Lc Ipsum has been the

Click Me! For Details



Dr. Ayesha Khan
Senior Consultant

Lorem ipsum dolor sit amet
consectetur adipisicing elit.

Cardiologist

Details



Dr. James Smith
Medical Officer

Lorem ipsum dolor sit amet
consectetur adipisicing elit.

Dermatologist

Details



Dr. Lisa Brown
Assistant Professor

Lorem ipsum dolor sit amet
consectetur adipisicing elit.

Neurologist

Details



Dr. Ahmed Rahman
Surgeon

Lorem ipsum dolor sit amet
consectetur adipisicing elit.

Orthopediatritian

Details



Dr. Emily Watson
Resident Doctor

Lorem ipsum dolor sit amet
consectetur adipisicing elit.

Peadiatritians

Details



Dr. Robert Green
Professor

Lorem ipsum dolor sit amet
consectetur adipisicing elit.

Internal medicine

Details

Latest Blogs



5 Tips for a Healthy Lifestyle

Discover the best practices to
maintain a healthy life. Eat well,
exercise regul...

Read More



Understanding Mental Health

Mental health is as important as
physical health. Managing stress,
getting enoug...

Read More



Top 10 Foods for Heart Health

Eating the right food can significantly
improve heart health. Some of the
best f...

Read More



Benefits of Yoga for Stress Relief

Yoga can help you relax and improve
mental well-being. Regular yoga
practice hel...

Read More

Smartcare

Lorem, ipsum dolor sit amet
consectetur adipisicing ecessitatibus
minus vitae autem obcaecati nihil
doloribus!

Follow us



Company

Home
Contact us
Service
Blog

Support

Help Center
Privacy policy
Terms of Services
FAQ

Contact

✦ Tower-4,CDA
Avenue,Chattogram
☎ +8801298374853
✉ smartcare@gmail.com

Your Service Booking is Confirmed External Inbox x



sharmin150907@gmail.com

to me ▼

Service Booking Confirmed! 🎉

Dear Rahim,

We are happy to inform you that your booking for the service **CBC** has been **confirmed**.

Scheduled Date: June 28, 2025

Thank you for choosing our service. We look forward to serving you! 😊

If you have any questions, feel free to reach out to us anytime.

Warm regards,
Smartcare Team

Confirm Your Email External Inbox x



sharmin150907@gmail.com

Thanks for creating an account on our platform Click the below link to verify your account Confirmation Link : <http://127.0.0.1:8000/patient/active/MjM/cs4dkw-eee53429c3f012cfb4899447f3aa8d2a/>



sharmin150907@gmail.com

to me ▼

Thanks for creating an account on our platform

Click the below link to verify your account

Confirmation Link : <http://127.0.0.1:8000/patient/active/MjM/cs4dkw-eee53429c3f012cfb4899447f3aa8d2a/>

← Reply

→ Forward

Conclusion

The Hospital Management System makes appointment booking and doctor-patient communication easier and more efficient. It helps improve hospital workflow through a secure, automated, and user-friendly platform.